DATA SYSTEMS ADMINISTRATOR · ENTRY-LEVEL COURSE · MODULE DS.01.05

TROUBLESHOOTING 101

Instructor Speaker Notes — a per-slide guide to presenting the deck

These notes walk you through all 23 slides of the Troubleshooting 101 deck, in order. Every slide gives you what is **on screen**, a suggested script (**SAY**), interaction prompts (**ASK**), step-by-step live demonstrations (**DEMO**), and the misconceptions and pitfalls to catch (**WATCH FOR**). The SAY lines are starting points, not a teleprompter — put them in your own voice, but keep the demos live and the drills interactive.

HOW TO READ THE LABELS

SAY — suggested talking points; paraphrase freely.

ASK — questions to pose to the class, with the answers you're listening for.

DEMO — do it live on the projector; slides alone don't teach tools.

WATCH FOR — misconceptions beginners bring in; surface them before they harden.

Pacing plan — about 150 minutes of class time

Slides	Segment	Time	Mode
1–2	Welcome + course route	~10 min	Talk
3–6	Unit 01 — Think like a troubleshooter	~25 min	Talk + drill
7–11	Unit 02 — Read the OSI ladder	~30 min	Talk + demo + game
—	Break	10 min	—
12–14	Unit 03 — Drive the Run menu	~20 min	Demo + type-along
15–18	Unit 04 — Triage with Windows tools	~30 min	Demo-heavy

Slides	Segment	Time	Mode
19–20	Unit 05 — Tune Internet Options	~15 min	Talk + demo
21–22	Unit 06 — Log it at the help desk	~15 min	Talk + example
23	Field kit + assignment	~5 min	Close

Splitting it? Two 75-minute sessions work well: Session A = slides 1–11 (method + OSI), Session B = slides 12–23 (tools, browser, logging).

Before class — setup & props checklist

- Instructor machine + projector, with **Command Prompt font enlarged** (students in the back must read every ping).
- A working printer reachable from the instructor machine — required for the **Print Spooler demo** (slide 17). Test it the day before.
- A pre-staged **bogus proxy entry** on the demo machine (Connections → LAN settings → check “Use a proxy server”, point it at a dead address) for slide 20.
- A spare **patch cable** as a physical prop for Layer 1 — pass it around.
- Whiteboard or flip chart for collecting student answers (slides 3 and 6).
- Printed **student handouts** and blank log-book pages — the Unit 06 assignment starts this week.
- Lab PCs on for the type-along in Unit 03; verify students can open the Run box.

WELCOME + UNIT 01

Slides 1–6 · Opening and Think Like a Troubleshooter

SLIDE 01 · COVER — "FIND IT. FIX IT. WRITE IT DOWN." · ~5 MIN

ON SCREEN Course title and the six-word tagline.

SAY

"Welcome. This course teaches one job skill: finding and fixing computer and network problems methodically — and then writing down what you did."

"The tagline is the whole job in six words: find it, fix it, write it down. By the last unit, that third part will be the habit that makes you employable."

ASK THE CLASS

"Who has ever fixed a tech problem at home? What did you do first?" — collect two or three answers. Someone will say "I restarted it" or "I guessed": perfect. Reply: "Today we replace guessing with a method."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Some students expect a theory lecture. Promise out loud that every unit has a live demo — and keep that promise; the demos are the course.

SLIDE 02 · COURSE ROUTE — ONE SKILL, SIX STOPS · ~5 MIN

ON SCREEN The six units as six stops, plus the "After this course" outcome box.

SAY

"Here's the whole journey on one slide. Six stops, and each stop hands you one tool or one habit."

"Read the outcome box with me: by the end you can isolate a fault layer by layer, open any admin tool in seconds, read the logs, and write the ticket. That's what a first-line tech actually does all day."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Don't linger here — it's a map, not a lesson. Thirty seconds per stop, then move; each unit explains itself when you get there.

SLIDE 03 · UNIT 01 OPENER — "THE INTERNET IS BROKEN" · ~3 MIN

ON SCREEN Unit title plus the classic vague ticket.

SAY

"A user walks up and says: the internet is broken. Where do you even start?" — then stop talking. Let the silence work for five seconds.

ASK THE CLASS

"What would you do first?" — write every answer on the whiteboard without judging. Most will be guesses: reboot, check Wi-Fi, call someone.

Then say: "Look at this board — every answer is a different guess. This unit replaces guessing with a method that always has a next step."

SLIDE 04 · THE SIX-STEP METHOD · ~10 MIN

ON SCREEN The six steps with one-line descriptions; footer notes it's the CompTIA-style industry method.

SAY

"Six steps. Same order. Every single ticket. This is the same loop real help desks run — learn it once and it works on every problem you'll ever meet."

Step 1 — Identify: "Gather facts before you touch anything. What exactly fails? Since when? What changed?"

Step 2 — Theory: "Start with the most likely cause, not the most exotic. If a PC won't turn on, check the power cable before you suspect the motherboard. Techs call this questioning the obvious."

Step 3 — Test: "Change ONE thing at a time. Change three things and the problem vanishes — you'll never know what fixed it, or whether it'll stay fixed."

Step 4 — Plan: "Restarting a service just happens. Reinstalling a driver gets a plan: what could break, and how do you roll back?"

Step 5 — Verify: "The original problem is gone AND nothing new broke — and the user confirms it, because their definition of fixed is the one that counts."

Step 6 — Document: "Write it down. Unit 06 turns this into a habit."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Beginners skip step 3 (one change at a time) and step 6 (document) the moment pressure hits. Flag both now by name — you'll grade them later, in the log-book assignment.

SLIDE 05 · TOP, BOTTOM, OR THROUGH THE MIDDLE? · ~6 MIN

ON SCREEN Three strategies: top-down, bottom-up, divide-and-conquer, each with a "best when".

SAY

"Once you have the facts, pick a direction. All three reference a ladder from the cable at the bottom to the app at the top — next unit gives you that ladder."

"Top-down: start at the app, work toward the cable — best when one app misbehaves and everything else is fine. Bottom-up: start at the cable and link light — best when nothing works at all. Divide-and-conquer: start in the middle with a ping test and split the problem in half — the pro's default when you're not sure."

ASK THE CLASS

Class call-out (as the slide prompts): "I describe a symptom — you call the direction, and why." Three rounds:

"Only Outlook fails, everything else works" → top-down (one app). · "A whole room loses connectivity" → bottom-up (total outage). · "One PC, vague 'slow internet'" → divide-and-conquer (start with ping).

SLIDE 06 · DRILL — HOW BIG IS THE BLAST RADIUS? · ~8 MIN

ON SCREEN Four scope questions, the practice scenario (3 of 30 PCs can't print; switch replaced yesterday), and the completion criterion.

SAY

> *"Before you touch anything, ask four questions: one machine or the whole room? One app or every app? Wired, Wi-Fi, or both? Did it ever work — what changed?"*
>
> *"Narrowing the scope is the core diagnostic skill. It turns 'everything is broken' into a problem you can actually test."*

LIVE DEMO — DO IT ON THE PROJECTOR

1. Run as think-pair-share: read the scenario aloud, pairs get two minutes.
2. Listen for the completion criterion: one sentence naming what IS affected (3 PCs, printing only), what is NOT (the printer — 27 PCs print fine; you changed nothing), and the FIRST test.
3. Model answer: the fault lives between the three PCs and the switch — and the switch is the last thing that changed. First test: check link lights and cables on those three PCs and their switch ports (Layer 1), then confirm the ports are in the right VLAN (Layer 2).

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Expect "reinstall the printer driver on all 30 PCs." Push back with the scope: why fix 27 machines that work? Fleet-wide actions on a three-machine problem is how technicians break things that weren't broken.

UNIT 02 + UNIT 03

Slides 7–14 · The OSI Ladder and the Run Menu

SLIDE 07 · UNIT 02 OPENER — READ THE OSI LADDER · ~2 MIN

ON SCREEN Unit title: four layers explain almost every network problem.

SAY

“The OSI model is a map, not memorization homework. Seven layers exist, but layers 1 to 4 are where nearly every hands-on fault lives — so that's where this course stays.”

“Learn the ladder once, and every symptom gets a home.”

SLIDE 08 · THE LADDER, BOTTOM TO TOP · ~8 MIN

ON SCREEN L1–L4 with what lives at each layer and one check question per rung; rule: start at L1.

SAY

“Present it bottom-up — L1 is where you start, and you climb only when the rung below checks out.”

L1 Physical: “Cables, NICs, Wi-Fi radio, link lights. The check question: is it plugged in? Is the link light on?” — hold up the patch cable prop.

L2 Data Link: “Switches, MAC addresses, VLANs. Can devices on the same switch see each other?”

L3 Network: “IP addresses, routers, DNS lookups. Can you ping the gateway? Does a name resolve to an IP?”

L4 Transport: “TCP, UDP, ports, firewalls. Is the app's port open — or is a firewall blocking it?”

ASK THE CLASS

“Why start at the bottom?” — listening for: each rung depends on the one below. A dead cable kills everything above it, so checking apps first wastes time.

SLIDE 09 · EVERY SYMPTOM HAS A HOME LAYER · ~6 MIN

ON SCREEN Symptom → layer map, L1 through L5+; class-drill prompt at the bottom.

SAY

> *“Read the map once together, then we play it as a game. No link light? L1. Same-switch devices can't see each other? L2. Can't ping the gateway? L3. One app blocked? L4.”*
>
> *“And this last row — ping works but browsing fails — lives above Layer 4. Park it. Unit 05 is where that problem goes.”*

ASK THE CLASS

Rapid-fire game: you call a symptom, the class calls the layer. Four or five rounds, pace quick.

Good extra symptoms: “Wi-Fi radio toggled off” (L1) · “PC in the wrong VLAN” (L2) · “Names won't resolve but IPs ping” (L3/DNS) · “Firewall blocks the print app” (L4).

SLIDE 10 · LIVE DEMO — FOUR PINGS TELL YOU WHERE IT BROKE · ~10 MIN

ON SCREEN The four pings with what each tests and what its failure means; companion tools at the bottom.

SAY

> *“One command bisects the problem space — this is divide-and-conquer from Unit 01 made concrete. The rule: the first ping that fails names the layer.”*

LIVE DEMO — DO IT ON THE PROJECTOR

1. Open cmd on the projector (font already enlarged). First run ipconfig — show students where to find the gateway address; they need it for ping 2.
2. ping 127.0.0.1 — "your own network stack; if THIS fails, the PC itself is broken."
3. ping <gateway> — "your local network; fails → cable, Wi-Fi, switch, or IP settings."
4. ping 8.8.8.8 — "the internet by raw IP; fails → router, ISP, or gateway config."
5. ping google.com — "adds DNS; fails while 8.8.8.8 passes → a DNS problem, not the network."
6. Narrate a failure reading even if all four pass: "If 1–3 had passed and 4 failed, what would be broken?" — DNS.
7. Mention the companions: tracert shows the path hop by hop; Wireshark watches real packets — advanced demo for later in the course.

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Students memorize 8.8.8.8 without knowing why. Explain it once: it's just a public DNS server with an address that's easy to remember — any reachable internet IP works for step 3.

SLIDE 11 · MYTH CHECK — TWO MYTHS TO DROP TODAY · ~4 MIN

ON SCREEN Myth 01: OSI is what's actually running. Myth 02: resetting the browser fixes network problems.

SAY

> *"Myth one: OSI is not what's running on the wire — it's a shared map for talking about networks. The internet actually runs on the simpler TCP/IP model. Use OSI as your troubleshooting map, not a photograph of reality."*
>
> *"Myth two: a browser reset only touches settings above Layer 4. It cannot fix a cable, switch, IP, or port problem. Run the ping ladder first — then tune the browser, in Unit 05."*

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Both misconceptions are extremely common with beginners — and with their relatives, who will ask them to "fix the Wi-Fi" by reinstalling the browser. Surface them explicitly now, before they harden into habit.

SLIDE 12 · UNIT 03 OPENER — DRIVE THE RUN MENU · ~2 MIN

ON SCREEN Unit title: Win+R, the fast lane to every admin tool.

SAY

"Pros don't hunt through menus — they type three letters and they're in. Watch."

LIVE DEMO — DO IT ON THE PROJECTOR

1. Before any explanation: press Win+R, type cmd, hit Enter — console open in two seconds. Do it once more slowly so they see the keys.
2. "That box is the fast lane to every admin tool you'll use this semester. The speed sells itself."

SLIDE 13 · SEVEN COMMANDS WORTH MEMORIZING · ~10 MIN

ON SCREEN The cheat sheet: cmd, msconfig, services.msc, eventvwr, devmgmt.msc, ncpa.cpl, inetcpl.cpl.

SAY

"Seven commands. Type each one with me as we cover it — muscle memory beats highlighting."

Walk the table: cmd — "ping, ipconfig, tracert live here." · msconfig — "startup and boot troubleshooting." · services.msc — "start, stop, configure services — next unit." · eventvwr — "the PC's own log book." · devmgmt.msc — "drivers and hardware; Layer 1's home on Windows." · ncpa.cpl — "adapters and IP settings." · inetcpl.cpl — "browser and proxy settings — Unit 05."

LIVE DEMO — DO IT ON THE PROJECTOR

1. Type-along: students open the Run box and launch each command as you name it; they close each window after a look.
2. Circulate while they type — the two most common typos are services.msc (missing the .msc) and ncpa.cpl (transposed letters).

ASK THE CLASS

"Which command opens the PC's log book?" → eventvwr. "Which one is Layer 1's home on Windows?" → devmgmt.msc.

Exit artifact: "Start your own cheat sheet this week — add every new Run command you meet, here or on the job."

SLIDE 14 · SAME TOOL, THREE DOORS + THE DECODER · ~8 MIN

ON SCREEN Three ways to open Event Viewer; the .msc / .cpl / .exe decoder; myth-busted footer.

SAY

> *"There are three doors into every admin console — and they all open the exact same tool. The Run box is simply the fastest door."*
>
> *"Now the endings stop being magic spells: .msc is a management console, .cpl is a Control Panel applet, .exe is a program."*

LIVE DEMO — DO IT ON THE PROJECTOR

1. Door 1: Start menu → type "Event Viewer" → open it.
2. Door 2: Win+R → eventvwr → Enter.
3. Door 3: in cmd, type eventvwr → Enter.
4. Point at all three windows side by side: "Same console. Three doors."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

The myth to bust: the Run box doesn't just launch programs — it opens management consoles and control panels too. And it still works when the Start menu itself is the thing that's broken, which is exactly when you'll need it.

UNIT 04 + 05 + 06 · CLOSE

Slides 15–23 · Windows Tools, Internet Options, Logging, Field Kit

SLIDE 15 · UNIT 04 OPENER — TRIAGE WITH WINDOWS TOOLS · ~2 MIN

ON SCREEN Task Manager, Services, Event Viewer — see what's running, what's stopped, what went wrong.

SAY

"Three tools answer three questions: what's running, what's stopped, what went wrong — in that order. Triage order matters: look before you kill."

SLIDE 16 · TASK MANAGER — WHO'S EATING THE PC? · ~8 MIN

ON SCREEN Processes tab, Startup tab, End task; baseline myth-check footer.

SAY

"Every 'my PC is slow' ticket starts here. Open it with Ctrl+Shift+Esc — or taskmgr in the Run box, like Unit 03 taught you."

"Processes tab: click a column header to sort by CPU, Memory, or Disk — the resource hog floats to the top. Startup tab: disable the high-impact apps that slow every boot — often the biggest free speed-up you'll ever deliver. End task: the safe way to kill a frozen app — after you've looked."

LIVE DEMO — DO IT ON THE PROJECTOR

1. Open Task Manager live; sort by CPU, then Memory, then Disk — narrate what floats up.
2. Show the Startup tab and read the impact column aloud.
3. Do NOT end a random task to be funny — say why: "Ending a task is the last step, not the first; unsaved work dies with it."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Baseline myth: high CPU isn't always bad. A busy PC isn't automatically a broken PC — updates, antivirus scans, and video calls all spike it legitimately. Know the machine's normal baseline first, then compare.

SLIDE 17 · SERVICES — THE APPS WITH NO WINDOW · ~10 MIN

ON SCREEN Startup types table, dependencies warning, and the Print Spooler class demo.

SAY

"A service is a program with no window — printing, updates, networking. Open services.msc, right-click a service, and you can start, stop, or restart it."

"Four startup types: Automatic — starts with Windows, always on. Automatic (Delayed) — shortly after boot, so boot stays fast. Manual — runs only when something asks. Disabled — never runs. Don't set Disabled if you're unsure."

LIVE DEMO — DO IT ON THE PROJECTOR

1. The Print Spooler proof (printer tested before class!): open services.msc, find Print Spooler.
2. Stop the service → try to print → it fails. Let the class see the failure.
3. Restart the service → print again → it works.
4. Land the point: "Nothing changed except the service. That's how you prove a cause instead of guessing one — step 3 of the method in its purest form."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Before stopping ANY service: check the Dependencies tab first. Stopping one service can quietly break three others that rely on it. The Print Spooler demo is the safe sandbox — zero risk, instant cause and effect.

SLIDE 18 · EVENT VIEWER — THE PC'S FLIGHT RECORDER · ~10 MIN

ON SCREEN The three logs that matter; how to read an event (Level, Event ID, Time, Filter).

SAY

"Windows writes down everything important — you just have to know where to read it. Open eventvwr, Windows Logs. Three logs matter: Application — app crashes. System — Windows itself: drivers, services, boot. Security — logons and access attempts."

"Reading an event: the Level tells you severity — skim to Error and Critical. The Event ID is the error's fingerprint — search it online and you'll find what it means. The Time lines events up with the incident. And the Filter hides the noise so you keep Errors only."

"To reconstruct an incident: open the log, filter to the time things went wrong, and read upward from the first Error. Logs are evidence — not noise — and you'll quote them in your log book in Unit 06."

LIVE DEMO — DO IT ON THE PROJECTOR

1. Open eventvwr → Windows Logs → System. Show the raw volume first: "This is why we filter."
2. Filter current log → tick Error and Critical → apply. Watch the noise disappear.
3. Open one Error event: point at Level, Event ID, Time. Search the Event ID live if you're online — students remember the day Google became a diagnostic tool.

SLIDE 19 · UNIT 05 OPENER — TUNE INTERNET OPTIONS · ~2 MIN

ON SCREEN inetcpl.cpl — where browser, cache, and proxy settings live.

SAY

"The network can be perfect while the browser still fails. This is the classic 'internet is down' ticket where the internet is fine."

"Sequence matters: run the ping ladder first. Only when all four pings pass do you look here."

LIVE DEMO — DO IT ON THE PROJECTOR

1. Launch it the Unit-3 way: Win+R → inetcpl.cpl → Enter. Flash the six tabs, then move on — next slide does the work.

SLIDE 20 · PING WORKS, BUT THE BROWSER DOESN'T · ~12 MIN

ON SCREEN The six tabs; the classic bogus-proxy ticket with its fix; the suspicion order and reset myth-check.

SAY

> *"Six tabs in one breath: General — homepage and cache. Security — trust zones. Privacy — cookies and pop-ups. Connections — LAN settings, where the proxy lives. Programs — default browser and add-ons. Advanced — deep settings and the big Reset button."*
>
> *"Browser problems live above Layer 4. If the ping ladder passes, suspect DNS, then proxy, then browser config — in that order."*

LIVE DEMO — DO IT ON THE PROJECTOR

1. The pre-staged bogus proxy: open the browser — every site fails. Run the four pings live — all pass. "Network healthy, browser dead. Where do we look?"
2. Connections tab → LAN settings → show the checked proxy, pointed at a dead address. "Sometimes malware leaves this behind."
3. Uncheck "Use a proxy server" → OK → reload the page → browsing returns. Let the class see both states.
4. Close the loop: "The proxy lives above Layer 4 — the network was fine the whole time; only the browser's settings were lying."

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Students love the Reset button because it feels decisive. It can't fix a cable, switch, or IP problem — it only rebuilds browser-level settings. Run the ladder first, reset second, and only when the problem genuinely lives in the browser.

SLIDE 21 · UNIT 06 OPENER — LOG IT AT THE HELP DESK · ~2 MIN

ON SCREEN The fix isn't finished until it's written down; step 6 becomes a habit.

SAY

> *"Remember step 6 of the method? This unit is where 'document' becomes a habit. The fix isn't finished until it's written down."*
>
> *"Frame it as a superpower, not homework: today's log entry is tomorrow's first-line fix. That's the knowledge loop every good help desk runs on."*

SLIDE 22 · A GOOD LOG ENTRY HAS SIX LINES · ~12 MIN

ON SCREEN The six lines; the annotated example entry; ticket fields; escalation etiquette.

SAY

> *"Six lines. WHO — who reported it and where, so anyone can find them again. WHEN — date and time, because that's what lines the story up with Event Viewer evidence. SYMPTOM — in the user's own words: 'Wi-Fi drops every few minutes' tells the next tech far more than 'network issue.'"*
>
> *"TRIED — every test and change, in order, including the dead ends; dead ends are data. RESULT — what each attempt did. NEXT STEP — fixed, monitoring, or escalate. Never leave the story without an ending."*
>
> *"Read the example entry line by line. Notice it quotes real evidence — the ping results. That's what separates a log from a diary."*
>
> *"A ticket system is a log book with search — same six lines plus fields: category, priority, status, resolution. And escalation etiquette: before you hand a ticket to tier 2, record everything you tried, so nobody repeats your work."*

ASK THE CLASS

"Why record the tests that failed?" — listening for: so the next tech doesn't repeat them; failures narrow the problem too.

"Why the user's own words for the symptom?" — translation loses detail; 'drops every few minutes' is a pattern, 'network issue' is not.

SLIDE 23 · YOUR FIELD KIT — THE WHOLE COURSE ON ONE CARD · ~5 MIN

ON SCREEN Four takeaways: the 6 steps, the ping ladder, the run commands, the 6-line log; the log-book assignment.

SAY

> *"That's the whole course on one card: a method, a ladder, a toolbox, and a habit. Everything else is practice."*

“Recite them with me: identify, theory, test, plan, verify, document — 127.0.0.1, gateway, 8.8.8.8, google.com — cmd, services.msc, eventvwr, inetcpl.cpl — who, when, symptom, tried, result, next step.”

“Assignment, starting this week: keep a log book at every lab. Use the six-line template for every problem you work — even the small ones. And keep growing that Run-command cheat sheet.”

“Find it. Fix it. Write it down. See you at the lab.”

WATCH FOR — COMMON MISCONCEPTIONS & PITFALLS

Don't skip the recitation to save two minutes — the chant is what they'll remember at 2 a.m. on a help-desk shift. Have the class say each list out loud with you, once, at full volume.

DS.01.05 · Troubleshooting 101 — Instructor Speaker Notes · 23 slides · Find it. Fix it. Write it down.